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# AI-assisted auditing and professional skepticism

## Auditors' perceptions of AI-generated audit evidence

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## Preface

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## Abstract

Artificial intelligence (AI) is increasingly being integrated into many industries, one of which is the auditing profession. In this context, AI-based tools are used to support data processing, anomaly detection and audit documentation. Although this technology may improve analytical capabilities and efficiency, it also reshapes the evidential environment in which audit evidence is both generated and evaluated. With this development within auditing, it also creates complexity for auditors in exercising professional skepticism toward AI-generated outputs. Hence, this study aims to provide a deeper understanding of auditors' perceptions of exercising professional skepticism when evaluating AI-generated outputs as part of audit evidence.

Through a qualitative research design based on semi-structured interviews with auditors, the collected empirical data was analyzed using thematic analysis. Professional skepticism continues to be perceived as a fundamentally human responsibility in AI-assisted auditing, where AI is understood as a supportive tool rather than a replacement for professional skepticism. At the same time, the integration of AI reshapes how professional skepticism is exercised, requiring auditors to evaluate both client-provided information, AI-generated information and the systems producing the outputs. The exercise of professional skepticism in AI-assisted environments is further influenced by factors such as professional experience, understanding of AI systems and organizational conditions including efficiency demands and time pressure. The study therefore contributes to an understanding of professional skepticism in AI-assisted auditing by conceptualizing it as a form of calibrated trust. This is where AI-generated outputs must be assessed critically, contextually and selectively before being integrated into audit decisions.

**Keywords**: Professional Skepticism, Artificial Intelligence, Audit Evidence, AI-assisted Auditing, AI-generated outputs, AI-generated information, Auditors' Perceptions

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# 1 Introduction

*This chapter introduces how the emergence of AI-assisted auditing is changing the conditions for exercising professional skepticism and explain why this change requires new research. The purpose and research questions conclude this chapter.*

## 1.1 Background

Professional skepticism is recognized as a crucial principle in auditing and serves as a safeguard for audit quality (Hurtt, 2010). According to the International Standard in Auditing (ISA) 200, auditors are required to have a questioning attitude and be critical when evaluating potential audit evidence (IAASB,2009). This involves critically assessing whether audit evidence is sufficient and appropriate throughout the audit engagement. Also in academic research, professional skepticism is conceptualized as both a questioning attitude and a strict evaluation of audit evidence (Nelson, 2009). Furthermore, skepticism is closely connected to how auditors interpret risk signals and make decisions under conditions of uncertainty according to Nelson (2009). With this foundation, Nolder and Kadous (2018) argue that professional skepticism can be grounded in attitude and mindset. This implies that skeptical actions are shaped by both the cognitive process and contextual factors, highlighting that skepticism is not a static trait.

The exercise of professional skepticism is being fundamentally reshaped by the emergence of Big Data, which significantly impacts how auditors' skepticism is exercised (Brown- Liburd et al., 2015; Yoon et al., 2015). Big Data, characterized by volume, variety, velocity and veracity creates according to Brown-Liburd et al. (2015) and Yoon et al. (2015) new layers of complexity; through the growing amount of unstructured data that traditional audit procedures cannot process effectively. Big Data gives auditors the availability of large and diverse datasets that can enhance their ability to find anomalies. However, both studies also emphasize that these extensive datasets also come with information overload that may obscure signals and hinder auditors from distinguishing reliable and relevant information from irrelevant cues. This highlights the growing complexity of the environment in which professional skepticism is exercised. In response to the complexities, auditors have turned to advanced technological solutions. Artificial intelligence (AI) elevates the analytical capabilities through AI-driven techniques that enable interpretation of large volumes of unstructured data; by filtering informational complexity while it identifies complex patterns and potential risk that traditional analytics would overlook (Yoon et al., 2015).

Building on these developments, more recent research investigates how AI changes the evidential environment. In this study, AI is explicitly defined following Kokina and Davenport (2017) who characterize it as systems that are capable of learning from data, identifying patterns and generating outputs that aid decision-making. Gu et al. (2024) further characterize AI as systems capable of extracting patterns from vast amounts of data and generate assessments that support and augment decision-making. In auditing AI systems include machine-learning models, used for risk assessment, predictive estimations and anomaly detection. These systems handle large amounts of unstructured data and generate outputs that assist in producing the evidential basis for audit decisions.

Prior research by Kokina and Davenport (2017) describes AI and automation as catalysts for a transition toward increasingly data-driven audit approaches and new analytical support. It shows that AI enables auditors to process and analyze extensive datasets and simultaneously identify patterns that may indicate anomalies or potential misstatements (Kokina and Davenport, 2017; Sun, 2019). More recent empirical research highlights the growing adoption of AI in audit firms and investigates possible associations with audit outcomes, including the audit quality (Fedyk et al., 2022). Yet, while the adoption increases, its effects on audit quality remain ambiguous, indicating that AI's impact is neither simple nor consistent. Concurrently, the integration of AI in everyday workflows presents both challenges and opportunities (Kokina et al., 2025). This raises questions about how auditors maintain professional skepticism, which is crucial for audit quality, while interacting with AI-based evidence. While Fedyk et al. (2022) focus on firm-level outcomes, Kokina et al. (2025) instead examine auditors' daily interactions with AI, suggesting that effects may differ between the organizational and individual levels.

Furthermore, machine-learning techniques such as predictive models have been shown to forecast the likelihood of future financial misstatements based on the historical financial data reported (Parker et al., 2025). When large datasets are transformed into structured AI outputs, these systems generate indicators such as predictive signals or anomaly alerts, supporting audit planning and evidential evaluation. Similarly to Sun (2019), Parker et al. (2025) extend the focus beyond anomaly detection to AI's evidential role in predictive capabilities.

In addition, Li et al. (2026) demonstrate that auditors' exhibit a significant decrease in skeptical behavior when using inherited AI tools compared to tools they have been involved in developing themselves. These findings suggest the exercise of professional skepticism is not merely influenced by technical performance but also by auditors' understanding, engagement and perceived control over tools. This differs from Fedyk et al.'s (2022) focus on the effects of adoption instead of the tool's origin which suggests that the origin matters and may influence auditors' trust and level of engagement. Similarly, studies have indicated that auditors' reliance on AI-based estimates is contingent upon contextual factors such as task complexity, level of uncertainty and algorithm adaptability (Commerford et al., 2022; Ulla & Commerford., 2025). Auditors' engagement in providing inputs to AI systems may increase perceived control, which can influence their reliance on AI outputs and thereby their use of professional skepticism (Commerford et al., 2024). These studies show that AI's impact on professional skepticism depends not only on the system's characteristics but also on the auditors' involvement.

Auditors can demonstrate algorithmic aversion, meaning they tend to place less trust in evidence produced by AI systems than identical evidence provided by professional auditors (Commerford et al., 2022). In contrast, a study by Logg et al., (2019) showed that users of AI may exhibit algorithm appreciation, meaning that the AI-outputs can sometimes be preferred over human auditors. These conflicting findings reveal a split in the literature as certain studies emphasize caution toward Al while others show over-reliance patterns.

Romeo and Conti (2026) draw attention to the risk of automation bias, where users put excessive trust in AI systems and fail to critically evaluate the outputs. This makes the concept of "co-piloted"

auditing important since it sheds light on AI outputs being insightful for auditors, while the responsibility for evaluating audit evidence and forming conclusions is still with the auditor (Gu et al., 2024). This highlights a fundamental tension where AI can support the auditors' decisions while at the same time challenging the critical mindset that supports professional skepticism, leaving ultimate responsibility with the auditor.

Research on AI use in auditing is not only about efficiency or control challenges but also ethical and governance issues. A systematic review raised concerns about algorithmic opacity and potential bias, emphasizing the importance of human oversight (Murikah et al., 2024). Other research further examines the normative and accountability challenges that are linked to decision-making based on AI outputs in auditing (Lehner et al., 2022). Compared to performance-oriented research, these ethical perspectives underscore risks that remain even when AI functions as intended. However, even with AI tools the auditor is ultimately responsible for professional decisions made and the outcomes according to Munoko et al. (2020). Together, they show that the duty to maintain professional skepticism does not diminish when AI tools contribute to generating or evaluating the audit evidence.

Beyond ethical and accountability concerns, AI not only affects auditors' responsibilities but also shapes how audit evidence is organized, prioritized and presented. Barr-Pulliam et al. (2024) discussed how AI fundamentally changes the conditions for generating and evaluating audit evidence by providing alternative methods for data collection and analysis. For example, it has become easier for auditors to move beyond limited data samples and efficiently analyze Big Data (Deloitte, n.d.). As a result, audit firms are making substantial investments in AI, including EY's 1.4$ billion investment in becoming an AI-powered professional service organization (EY, 2024). Similarly, PwC emphasizes the need for a human-led and tech-powered approach (PwC, n.d.); while KPMG highlights that AI enabled data-driven auditing through their own platform called KPMG Clara (KPMG, n.d.). Therefore, the ongoing AI-transformation within the auditing industry underscores the importance of developing a deeper understanding of how auditors understand and apply a skeptical mindset in AI-supported environments.

## 1.2 Problem discussion

Prior research on professional skepticism has largely developed in settings where audit evidence is produced through traditional procedures and direct client interactions and is primarily based on structured data (Nelson, 2009; Nolder & Kadous, 2018). The growing prevalence of AI-generated outputs introduces a newer and evolving evidential framework mainly characterized by the integration of large volumes of unstructured data, which challenges the ability to obtain adequate and relevant audit evidence via conventional audit approaches (Brown-Liburd & Vasarhelyi., 2015; Li et al., 2025). This indicates that with the increasing complexity and reduced direct interpretability of audit evidence due to the combination of unstructured data besides the structured data, auditors face greater expectations of heightened skeptical actions when assessing this evidence. Despite the significance of this emerging dual skeptical role, prior research has only begun to explore these dynamics; and therefore understanding them is relevant to grasp the difference between these forms of audit evidence and how they shape the auditors' skeptical role.

While existing literature has effectively shed light on AI adoption, system performance and the dynamics of auditor reliance in AI-assisted audit settings, it has primarily examined how auditors adjust their reliance on AI-based decision support systems in response to contextual and algorithmic conditions, providing deeper understanding of the calibration of reliance decisions (Commerford et al., 2024; Ulla & Commerford, 2025). However, the research is primarily focused on measurable results like reliance behavior, predictive precision and audit efficiency (Fedyk et al., 2022; Rahman et al., 2024) and thereby offers comparatively far less insight into auditors' interpretation of the skeptical task when AI-generated outputs constitute part of the audit evidence.

This lack of knowledge is crucial as professional skepticism implies more than merely whether auditors rely more on a particular source of information. Rather, it involves how auditors interpret evidential cues/indicators, evaluate the reliability and relevance of evidence and perceive their duty to critically assess whether the evidence is sufficiently convincing to substantiate audit conclusions within distinct informational contexts (Nelson, 2009; Nolder & Kadous, 2018). As AI-based tools increasingly transform how evidence is generated, organized, and presented, auditors' subjective interpretations of these outputs become central to grasping the actual practice of skepticism (Barr-Pulliam et al., 2024; Li et al., 2026). Compared to reliance and performance-oriented studies, this highlights an interpretive dimension of skepticism that is still underexplored.

Comprehending the perceptions of auditors is therefore important for both conceptual and practical considerations, but also because perceptions play a crucial role in forming auditors' attitudes toward AI adoption and their willingness to rely on AI-generated outputs. It is especially important for advancing knowledge about how professional skepticism is interpreted in AI-assisted evidential contexts. Evaluating AI-generated outputs may involve more than simply determining their reliability, it may also require auditors to reflect on the perceived trustworthiness of the algorithmic signals, the challenges of transparency and the understanding of accountability (Murikah et al., 2024). Exploring perceptions thus uncovers such aspects that are not captured by studies that focus on reliance or technical performance.

Empirically, exploring auditors' perceptions provides new knowledge about how practitioners comprehend and navigate AI-generated evidence in real-world audit contexts. While experimental studies have usefully identified conditions under which auditors may over- or under- rely on AI-generated outputs (Commerford et al., 2022; Ulla & Commerford, 2025), which are consistent as well with studies on human-algorithm interaction that shows that humans have tendencies to exhibit behaviors such as automation bias and algorithm aversion when engaging with algorithmic systems (Dietvorst et al., 2015; Logg et al, 2019; Romeo & Conti, 2026). However, fewer studies explicitly examine professional skepticism or its interpretation and exercise by auditors in practice. By drawing on auditors' own narratives, this study provides insights into how they evaluate AI-generated outputs, what they perceive as challenging in such evaluations and their skeptical duties in the presence of such outputs in real audit engagements; an aspect that prior research has only addressed through patterns of reliance or technical performance.

These advancements underscore the growing need to better understand auditors' perceptions and interpretations of professional skepticism especially in contemporary audit settings where evidence is increasingly shaped by AI-based tools.

## 1.3 Aim & Research Questions

The purpose of this study is to explore how auditors perceive the exercise of professional skepticism when evaluating AI-generated outputs as part of audit evidence.

To address this aim, the study is guided by the following research questions:

- How do auditors perceive the factors influencing professional skepticism in AI-assisted audit environments?
- How do auditors perceive the implications of AI for their skeptical responsibilities?
- What challenges do auditors perceive in maintaining professional skepticism when evaluating AI-generated outputs?

# 2 Conceptual and theoretical framework

*The chapter presents the central conceptual and theoretical frameworks that explain how AI reshapes the evidential environment, how skepticism functions as a mechanism, and how human-algorithm interaction affects the skepticism.*

## 2.1 Artificial intelligence and the transformation of the evidential environment

Recent research in the auditing field has progressively explored the implications of AI for the audit practice, especially regarding how these algorithmic systems impact the generation and assessment of audit evidence (Fedyk et al., 2022; Kokina & Davenport, 2017). Rather than simply boosting computational performance, AI-driven tools may change the informational settings in which auditors analyze financial data and detect possible misstatement risks.

Early research highlighted the potential of AI-enabled analytical tools to enhance auditors' ability to examine complex datasets and identify financial misstatements patterns. Kokina & Davenport (2017) illustrate how such emerging tools facilitate the transition from conventional sampling approaches to more thorough, data-driven analyzes. By processing extensive datasets and identifying interrelations among variables, AI-based tools generate signals that assist auditors in detecting potential irregularities in financial data. This study draws on these insights to exemplify how AI widens the character of evidential indicators that auditors are required to interpret.

Empirical studies show that AI-based systems are more prevalent in audit practice. According to Fedyk et al. (2022), the integration of AI within audit firms is linked to changes in audit procedures and outcomes, including enhanced analytical capabilities and operational efficiencies. Similarly, Rahman et al. (2024) establish that audit firms and their clients' use of AI may impact both audit quality and efficiency. Additionally, Appelbaum et al. (2017) point out that AI enhance the variety of evidential signals accessible to auditors, facilitating the discovery of anomalies and patterns that might be overlooked by traditional procedures. These findings support that AI-based systems are progressively becoming a part of the digital infrastructure that supports contemporary auditing, while highlighting various elements of this development from efficiency improvements to restructuring the evidential scope.

A key application of AI in auditing is machine learning models, that identify financial reporting risks or accounting misstatements. Bertomeu et al. (2021) illustrate that machine learning can identify associations that correspond to misstatement trends. Likewise, predictive models based on machine-learning approaches have been shown to forecast future material misstatements, offering a forward-looking perspective on risk assessment (Parker et al., 2025). Such models show how AI-based systems can produce indices that help auditors in identifying high-risk areas. Research on deep learning further underscores the potential impact of AI on audit processes. Sun (2019) provides a framework showing how deep-learning tools can be used on audit data to identify intricate patterns within financial transactions. The outputs of such models, often as anomaly indicators can reveal transactions for further investigation that might not be easily identified through manual ~~and~~ examination.

The significance of these developments extends beyond speed or analytical gains. Several studies highlight the role of AI tools in improving auditors' capabilities in managing and deriving insights from large datasets (Appelbaum, 2017; Kokina & Davenport, 2017). In parallel, other research suggests that these technologies also alter the way information is interpreted and assessed. Brown-Liburd et al. (2015) find that AI-enabled Big Data analytics and AI-mediated outputs can impact how auditors process and prioritize information, especially in complex judgment and decisive scenarios. This suggests that AI not only increases the volume of available data but also the perception of evidential cues. This is important for this study to understand how AI may influence auditors' evaluative process, especially in relation to professional skepticism.

More broadly, AI and digital advancements are actively changing the foundational nature of audit evidence as a growing array of studies indicates. Barr-Pulliam et al. (2024) assert that innovations influence how evidence is generated and assessed, reshaping evidential dynamics of the contemporary audit. Formerly, evidence has been gathered through methods such as document review, third-party confirmations and management inquiries, where auditors had unmediated contact with financial records and documents to evaluate the sufficiency and reliability of evidence. Meanwhile, an extra layer of algorithmic intermediation is added to the evidential environment due to the rise of adoption of AI systems. As AI generates outputs such as risk scores or flagged anomalies, auditors are required to assess evidence that has been pre-processed with AI-based systems. These computational outputs represent indicators that could inform auditors' evaluation of financial reporting risks. Thus, AI-driven systems yield analytical outputs that may serve as evidential signals in the audit process (Bertomeu et al., 2021; Parker et al., 2025). Rather than functioning as direct evidence in the traditional sense, these outputs can be perceived as analytically produced indicators that still demand further interpretation and validation. This illustrates a shift from auditors' direct involvement with the underlying data towards engagement with system-generated portrayals of that data, which is central to understand in this study.

Further research on the behavioral aspects of audit and analytical tools provides more depth. Li et al. (2026) indicate that the source of an analytical tool matters when auditors rely on them; auditors exhibit different levels of skepticism when they inherit a technique versus when they construct it independently. Their findings show that transparency and documentation about a tool can reduce potential negative effects on skeptical evaluation. This shows that the changes in the evidential picture are not only a technological evolution but also an interpretive one, as auditors evaluate both the output and the mechanisms leading to them. Yet, current research has not fully addressed how auditors navigate these interpretive challenges when engaging with AI-based tools.

## 2.2 Professional skepticism: normative expectations and conceptual insights

### *2.2.1 ISA standards on professional skepticism*

Professional skepticism is understood as an approach by auditors when evaluating audit evidence. In particular, ISA 200 stipulates that auditors must maintain a questioning mindset and critically evaluate whether the audit evidence is sufficient and appropriate. According to ISA 200, paragraph 13, professional skepticism is defined as an attitude entailing a questioning mind, a heightened awareness of the circumstances that may indicate possible misstatements that arise from error or

fraud (IAASB, 2009). Additionally, ISA 200, paragraph 15, A21, imposes a clear requirement on auditors to incorporate professional skepticism in different phases of the audit, acknowledging the possible presence of material misstatements in the financial statements (IAASB, 2009). This implies that professional skepticism is not restricted to a part of the audit but is expected to be maintained consistently beyond the planning or execution of the audit. Its purpose is to reduce audit risks encountered during the audit process such as failure to recognize unusual events, overgeneralizing conclusions drawn from limited evidence; which require auditors to critically assess the audit evidence. In particular, when inconsistencies arise where evidential information that contradicts other findings or raises concerns about the credibility of documents need to be assessed. In ISA 200, paragraph 15, A24 mandates that auditors perform additional investigation with sustained skepticism when uncertainty is present, even when the management has shown honesty and integrity (IAASB, 2009).

ISA 240 reinforces this with focus on fraud in financial statements audits. ISA 240, paragraph 6, highlight that fraudulent reporting often involves deliberate and intentional concealment strategies (IAASB, 2009). Therefore, ISA 240, paragraph 16, stipulates that audit team discussions in connection with audit engagement should disregard assumptions about management's honesty and integrity (IAASB, 2009). This enhances their ability to identify areas of financial statements that may be susceptible to material misstatement due to fraud. Furthermore, ISA 315, paragraph 13, underscores the significance of maintaining objectivity in audit evidence collection, where auditors are required to consider confirming and contradictory evidence when evaluating risks (IAASB, 2019). This becomes important in complicated situations where management bias is more likely to occur. Consequently, maintaining professional skepticism guarantees that audit evidence is thoroughly evaluated and is a reliable basis to identify material misstatements.

### *2.2.2 Conceptual perspectives on professional skepticism*

In academic literature, professional skepticism has been further developed through theoretical frameworks. It has been conceptualized as a combination of cognitive orientation and evaluative actions during the auditing process. A prominent framework by Nelson (2009) distinguishes professional skepticism into two components, skeptical actions and skeptical decisions. The skeptical actions refer to auditors' procedures to gather audit evidence, while skeptical decisions are assessing the probability that a misstatement may exist. The link between these two components is that the auditor's decision guides their behavior and translates into concrete audit actions (Nelson, 2009). Simultaneously, Nelson's (2009) framework also emphasizes that the exercise of professional skepticism is influenced by contextual factors such as incentives, evidential cues and the broad informational environment. The decisions and actions are influenced by the characteristics of the auditors, including personal traits, knowledge and incentive. The personal traits that could influence could be confidence and a tendency to question while knowledge is the auditors' experience and their ability to recognize patterns of misstatement. However, the various incentives refer to both social and economic influences that shape auditor behavior. It includes concerns about litigation risk and damage to reputation which could promote professional skepticism. On the other hand, constraints for instance desire to preserve a client relationship or time pressure and limited budget, can suppress the level of professional skepticism applied (Nelson,

2009). Moreover, evidential cues represent the diverse set of information collected and considered during the audit, which serve as the foundation for auditors' decisions. This information combined with auditors' prior experience and training; shape the informational environment in which evidence is continuously assessed. Within this process, actions generate new evidence which in turn revise subsequent decisions and influence future decisions, creating a continuous process (Nelson, 2009).

Similarly, Nolder and Kadous (2018) conceptualize professional skepticism as a questioning mindset together with attitude through active engagement in the evaluation of evidence. This perspective highlights that skepticism involves interpreting evidential signals and seeking additional evidence during times of uncertainty. A questioning mindset refers to how an auditor thinks, a deliberate stance characterized by objectivity, alertness and openness to signals of bias, fraud or other evidential inconsistencies. This way of thinking helps ensure the auditor remains open to new information rather than adopting a routine of "checking the boxes". At the same time, active engagement stands for the cognitive processes through which auditors thoroughly analyze audit evidence. This engagement includes conducting extensive and informative searches for evidence that may dispute management's claims, while generating and evaluating other explanations of the findings (Nolder & Kadous, 2018)

Although theoretical frameworks regarding professional skepticism emphasize that it is influenced by contextual factors, empirical research indicates these influences to be apparent in practice. In particular, studies have shown that the exercise of professional skepticism is shaped by factors such as time pressure, task complexity and the type of evidence available (Barr-Pulliam et al. 2024; Nelson, 2009). This indicates that the professional skepticism being pushed by standards to be applied is not being applied uniformly in practice because auditors' skepticism varies depending on the situation.

The growing use of AI has created new obstacles for the exercise of professional skepticism. When outputs are generated by AI systems, the auditors need to evaluate how they should be interpreted and incorporated into audit assessments. Instead of directly identifying anomalies through manual examination of financial data, the auditors will encounter structured outputs generated by AI.

In this AI-assisted auditing context, Nelson's (2009) framework of skeptical judgment and skeptical action becomes particularly salient. Skeptical judgement captures auditors' cognitive assessment of the reliability and relevance of AI outputs, while skeptical action reflects the extent to which auditors pursue further audit procedures such as seeking corroborating evidence for the system's results. Consequently, the integration of AI in the auditing process may reshape auditors' cognitive evaluations and behavioral responses to AI-generated outputs, depending on their perceived reliability of and trust in algorithmic outputs.

Recent work by Gu et al. (2024) provided an example of AI assisting in an auditing environment called Co-piloted auditing. In this context, AI systems are understood as decision-support tools for auditors, providing insights to assist in detecting financial reporting risks. Nonetheless, these systems function as a decision-support tool, and skepticism must still be exercised by auditors. Whether the evidence has been detected by the human eye or AI tools, the auditor is still

responsible for the interpretation and conclusions based on the information provided. The concerns are not limited to AI in auditing but also the ethical and governance implications of it. Consequently, the importance of human oversight is highlighted when AI contributes to professional decision-making when incorporated into the evidential process. The research has underscored the potential concerns for bias and accountability thereby reinforcing the importance of critically evaluating AI-generated outputs (Lehner et al., 2022; Munoko et al., 2020; Murikah et al., 2024).

## 2.3 Behavioral mechanisms: under-reliance and over-reliance

### *2.3.1 Conceptual perspectives on algorithm interaction*

Exercising professional skepticism necessitates that auditors thoroughly examine all sources of audit evidence (IAASB, 2009). However, research on human-algorithm interaction demonstrates that individuals typically exhibit responses molded by systematic behavioral tendencies (Parasuraman & Riley, 1997). Two tendencies that were frequently highlighted in the literature are automation bias and algorithm aversion.

Algorithm aversion describes the tendency to distrust and avoid algorithmic recommendations once observing an error, even when the algorithm's performance objectively exceeds human judgment (Dietvorst et al., 2015). In their experiments, Dietvorst et al. (2015) demonstrate that even minor errors lead individuals to lose confidence in algorithmic tools and prefer instead human judgment, regardless of the statistical evidence that confirms the algorithmic superior accuracy. This implies that even when overall performance is elevated, trust in algorithmic systems may be vulnerable to apparent errors which illustrates a central characteristic of this tendency as well as a "double standard" where comparable or even more serious human mistakes are more tolerated.

This phenomenon arises in forecasting and predictive contexts about future outcomes where users are directly acquainted with the tool's performance and effectiveness (Dietvorst et al., 2015). Experts and experienced individuals are most likely to be susceptible to algorithm aversion, often because they view algorithmic outputs as a threat to their professional judgment or believe that humans has a distinctive capacity to learn from mistakes and improve through experience which algorithms and machines lack (Dietvorst et al., 2015).

In contrast to algorithm aversion, automation bias refers to the tendency to excessively trust automated systems and fail to sufficiently question their outputs (Logg et al., 2019; Romeo & Conti, 2026). It often emerges when automated tools are perceived as technically sophisticated, objective, and analytically superior (Parasuraman & Riley, 1997). In such contexts, individuals may accept algorithmic recommendations without thoroughly evaluating the assumptions, constraints or mistake potential that are inherent in the systems. Skitka et al. (1999) show that individuals may comply with automated guidance despite its inaccuracy, indicating that these automated outputs can influence decision-making behavior. This tendency is related to Logg et al.'s (2019) research on algorithmic advice and especially the term of algorithm appreciation referring to that individuals may exhibit greater trust in algorithmic recommendations compared to human advice in specific situations. Although related, algorithm appreciation is conceptually distinct from automation bias

as it denotes an inclination towards algorithmic suggestions rather than a deficiency in critically evaluating them.

Automation bias is primarily characterized by two error types: omission errors where individuals fail to act when the automated system gives no guidance, and commission errors where individuals follow incorrect automated advice despite access to fully reliable, contradictory information. This bias frequently emerges in context of significant workload, time pressure or complicated multitasking, wherein individuals tend to act as a "cognitive miser" and depend on automation as a heuristic to reduce cognitive effort (Romeo & Conti, 2026; Skitka et al., 1999). The most vulnerable individuals are novices and individuals with limited to moderate AI literacy; who tend to overestimate their competence and understanding of the system, along with professionals who lack the domain-specific expertise needed to evaluate independently automated outputs (Romeo & Conti, 2026). Without performance feedback, individuals tend to display "algorithm appreciation" favoring algorithmic advice over human judgment which typically persists until they witness the algorithm make an error (Logg et al., 2019), which reflects a distinct mechanism than automation bias, as it concerns an initial preference for the algorithmic advice rather than misinterpreting its reliability.

These perspectives indicate that responses to these systems are diverse and potentially involve both over-reliance and under-reliance based on how the system and the context of where it is used, as well as the visibility of error and the user's own confidence are perceived. These concepts help to frame how auditors may interpret and respond to AI-generated outputs and how these influence their skeptical evaluation.

### *2.3.2 Empirical insights in auditing contexts*

To comprehend how these behavioral tendencies appear in auditing contexts, empirical research offers additional insights into auditors' interaction with AI-generated outputs. As AI-generated outputs are increasingly used to inform auditors' decisions, similar behavioral patterns can be observed in audit settings. These patterns are fundamentally important to auditors' exercise of professional skepticism as they influence their extent to question, challenge or verify algorithmic outputs.

Research by Commerford et al. (2022) and Ulla & Commerford (2025) examining auditors' interactions with algorithmic estimates indicates that auditors' reliance on AI-generated outputs is contingent on task complexity, uncertainty toward estimates and perceptions of the algorithms' flexibility. This implies that the degree of auditors' trust in algorithmic outputs is not constant but depends instead on their understanding of both the task and the system. When auditors perceive algorithms as inflexible and lack the ability to incorporate contextual knowledge, they may favor their own professional judgment discounting the algorithmic signal.

Moreover, the extent of auditors' involvement in constructing or calibrating analytical systems appears to moderate this effect. Commerford et al. (2024) find that when auditors engage in providing inputs and assessments to AI-based systems, their perceived control over the system rises which in turn affects their willingness to rely on algorithmic estimates, potentially reducing

adverse reactions. Romeo and Conti (2026) highlight that the structured and quantified nature of algorithmic outputs, such as risk indicators or predictive scores, can enhance the user's perceived technological authority and thereby bow to these signals. Reliance patterns on algorithmic tools depends not only on the perceived credibility of the advice source, but also on the degree of familiarity and the perceived control over these systems which might shape the usage of their outputs.

Drawing on broader and prior research into human-algorithm interactions, automation bias can be a distinct challenge in auditing (Commerford et al., 2022). This bias emerges when individuals regard algorithmic outputs as precise, data-driven, or technically advanced and therefore accept them as reliable without any adequate independent assessments (Logg et al., 2019; Parasuraman & Riley, 1997). In audit settings, this implies that auditors may accept AI-generated indicators of financial reporting risk as reliable due to their sophistication without any adequate independent assessments. On the other hand, algorithm aversion may lead individuals to underutilize the algorithmic outputs especially when the tools are perceived as rigid or subject to inaccuracies (Dietvorst et al., 2015), which could likewise hinder auditors' effective engagement with AI-based tools. Professional skepticism may decrease as a result just where it is most required. Algorithm aversion and automation bias are therefore two opposing but equally significant factors that threaten the exercise of professional skepticism in AI-assisted audit settings.

## 2.4 Theoretical synthesis

The previous sections have shed light on three interrelated parts that are central to the study, namely the increasing use of AI in the auditing process, behavioral biases in auditor decision-making and professional skepticism. These elements are closely connected within the contemporary audit environment and should not be understood in isolation. The implementation of AI in auditing has transformed the nature of the audit evidence environment by providing outputs that assist auditors in detecting risks and anomalies in financial data. Nevertheless, the increasing use of AI, driven by the growth of unstructured data, introduces behavioral tendencies that can influence how auditors engage with such outputs. The studies on algorithm aversion and automation bias suggest that auditors may either under-rely on the algorithmic insights or over-rely on them without adequate critical evaluation. As a result, the assessment of algorithmic outputs as audit evidence is not purely objective and can be shaped by auditors' behavioral responses. These processes interact in a reinforcing manner, whereby AI-generated outputs influence auditors' behavioral responses, which in turn determine the extent in which professional skepticism is exercised when assessing those outputs.

Therefore, within this transformed audit environment professional skepticism plays an important role as it enables auditors to critically assess algorithmic outputs. This entails questioning the outputs' reliability, considering alternative explanations and deciding whether additional audit procedures are necessary. Thus, it serves as a shield against the risk of auditors either uncritically accepting the AI outputs or readily dismissing potentially valuable insights. This interpretation is consistent with ISA 200, paragraph 11, which emphasizes that the auditor's overall objective is to obtain reasonable assurance that the financial statements are free from material misstatements

(IAASB, 2009). Auditors need therefore to effectively exercise professional skepticism by critically evaluating these AI outputs, without disregarding them too readily. Consequently, professional skepticism can be understood as a balancing mechanism through which the interaction between algorithmic outputs and auditors' behavioral responses is regulated.

Simultaneously, the exercise of professional skepticism is shaped by the auditors' perceptions regarding the transparency and credibility of the AI system, which may influence the interpretation and operationalization of professional skepticism. Given this, the effectiveness of professional skepticism in AI-assisted auditing depends not only on the auditing standards but also on auditors' cognitive responses to algorithmic signals. Hence, the theoretical framework integrates perspectives from AI-generated insights in auditing, behavioral biases and professional skepticism to elucidate how auditors interpret algorithmic signals and exercise skepticism in an increasingly data-driven audit environment. Figure 1 provides a simplified representation of the key relationships identified in this synthesis.

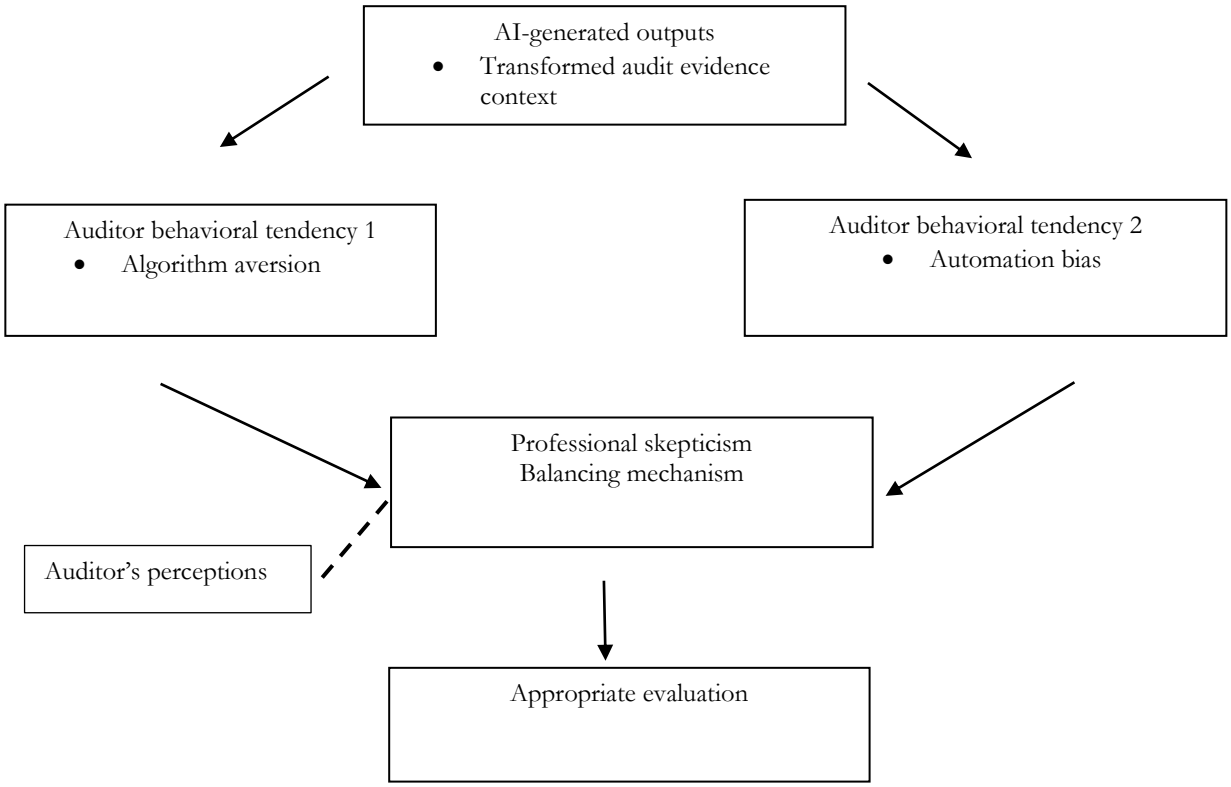



Figure 1- Interaction between AI outputs, behavioral tendencies and professional skepticism.

# 3 Methodology

*In this chapter, the selected approach is presented, from research strategy, sampling, data collection to the data analysis process, ethical considerations and trustworthiness.*

## 3.1 Research strategy

This study employs a qualitative research approach, as its intention is to examine how auditors understand and implement professional skepticism in audit contexts where AI-based systems contribute to the generation of audit evidence. Because skepticism entails interpretive judgment, a qualitative approach is suitable to examine how auditors understand and construct their skeptical assessments. Qualitative approaches are ideal when the objective is to examine how individuals interpret professional roles and practices in particular settings (David & Sutton 2011), which is consistent with this study's focus on auditors' reasoning and interpretations. Given the exploratory character of the aim and research questions, depth of insights was favored over generalizability which further justifies the adoption of a qualitative design.

The research process was guided by an abductive approach as it involves a continuous back-and-forth between the theoretical constructs on human-algorithm interaction and the evolving empirical insights. This continuous movement facilitates the refinement of the study's theoretical framework as patterns in auditors' reasoning emerged during the interviews.

Semi-structured interviews are also adopted in this study as they allow participants to express their own interpretations of experiences and practices while ensuring that relevant themes are covered, which makes them particularly appropriate for exploratory studies (David & Sutton, 2011; Denscombe, 2017). This design is appropriate to capture variation in how auditors understand and approach their work, as well as how they describe their professional responsibilities across different audit settings.

## 3.2 Sampling

The study adopts a non-probability sampling design, combining purposive and convenience sampling. This approach is suitable given the exploratory aim of examining how auditors perceive and exercise ~~p~~professional skepticism in contexts involving emerging AI-generated audit evidence. In qualitative research, such approaches are common when the aim is to get in-depth insights from information-rich cases rather than large and statistically representative samples (David & Sutton, 2011; Denscombe, 2017). Purposive sampling was employed to target authorized auditors who have experience with audit engagements involving AI tools. Their formal responsibility to evaluate audit evidence and exercise professional skepticism made them relevant for providing insights into the implications of AI on auditors' reasoning. Simultaneously, elements of convenience sampling were inevitable as access to auditors was influenced by their availability during the busy audit periods. Dependence on accessibility and practical constraints is typical in qualitative research and is frequently addressed as a pragmatic aspect of non-probability sampling (David & Sutton, 2011; Denscombe, 2017).

Participants are drawn mainly from within Big 4 firms, where the adoption of AI has advanced more significantly. The distribution among these firms is not uniform, indicating disparities in the extent to which each firm has adopted AI-based tools. This discrepancy is of analytical importance, as it allowed the study's ability to capture differing experiences of auditors based on their familiarity with AI. All participants are authorized auditors with substantial professional experience, positioning them well to reflect critically on the exercise of professional skepticism within increasingly AI-assisted audit environments. Recruitment is carried out until no new significant insights are revealed regarding auditors' accounts of skeptical evaluation in AI-assisted settings, suggesting that an adequate level of thematic saturation has been reached. This aligns with the principles of qualitative research where emphasis is on the richness and depth of the data rather than the size of the sample (David & Sutton, 2011). The objective is not to attain representativeness within the profession, but to obtain experience-based accounts from auditors well positioned to reflect on the role of skepticism in these audit settings. An overview of the participants, including job title, professional experience and interview length is presented in Table 1:

Table 1: Summary of respondents

| Participant | Job title | Years of professional experience | Interview length |
|---|---|---|---|
| P1 | Senior manager | 9 years | 59 min |
| P2 | Senior manager | 22 years | 36 min |
| P3 | Senior Manager | 9 years | 33 min |
| P4 | Manager | 5 years | 60 min |
| P5 | Partner | 18 years | 57 min |
| P6 | Director | 20 years | 41 min |
| P7 | Manager | 5 years | 58 min |
| P8 | Senior Manager | 21 years | 45 min |
| P9 | Partner | 31 years | 54 min |
| P10 | Manager | 5 years | 60 min |
| P11 | Director | 15 years | 33 min |
| P12 | Director | 14 years | 51 min |

## 3.3 Data collection

The data is collected through semi-structured interviews which, according to Denscombe (2017), are suitable for capturing the depth of perspectives and experiences of the participants. In relation to the aim of the study, this method is employed to explore how auditors interpret and enact professional skepticism in connection with AI tools.

Participants have been found through professional networks such as the FAR member list and LinkedIn but were later approached through email communication. The email provided participants an invitation outlining its purpose and participation conditions. It also included an information letter explaining confidentiality and anonymity, the voluntary nature of participation and the right to withdraw.

The interview guide is developed based on the study's research questions and structured into thematic areas. According to Denscombe (2017), interview guides provide structure to the

discussion while maintaining flexibility in how questions are asked. Following a pilot interview, overlapping or unclear questions were rephrased or removed, which improved clarity. The guide moved from general to more specific themes where participants could describe their professional roles before reflecting on more challenging aspects involving AI-generated outputs and skepticism.

Importantly, the interview design was constructed to capture perceptions of professional skepticism by asking participants to reflect on concrete audit situations where they assessed AI-generated outputs. This facilitated access to both cognitive and behavioral dimensions of skepticism. The questions encouraged them to discuss the reliability of such outputs, when they questioned them and how they actively engage in evaluating evidence and considering alternative explanations. Open-ended questions and follow-up probes supported detailed responses describing their own situations and experiences, thereby revealing how professional practices are carried out.

The interviews were conducted either digitally via Teams or in the respondent's office. The choice was based on considerations of accessibility, flexibility and the possibility of engaging participants across dispersed geographical locations within Sweden. This approach enabled participation of the authorized auditors despite their demanding work schedules. However, the interviews lasted between 30 and 60 minutes, with variation reflecting differences in participants' level of elaboration. Although some participants were more communicative than others, providing more extensive accounts and occasionally addressing topics beyond the initial questions or anticipated follow-up questions without probing; this variation did not negatively affect the quality or depth of the data collected.

All interviews are audio-recorded with participants' informed consent to ensure a precise capture of the data. This allows the researcher to focus on interaction rather than note-taking (Denscombe, 2017). The audio recordings are manually transcribed word for word, to transform spoken interaction into written text for analytical purposes to ensure that the analysis is grounded in the participants' own accounts.

Although interviews have the capacity to provide rich and in-depth qualitative data, they still present certain challenges. One such limitation is the risk of social desirability bias, where participants may align their responses in accordance with perceived professional norms rather than their actual practices (Bryman, 2016). To address this limitation, the interviews focused on allowing participants to reflect on their own specific instances and experiences rather than general statements, which strengthened the study's ability to capture how professional skepticism is enacted in real-life AI-assisted audit settings.

## 3.4 Data analysis

The empirical data were analyzed using thematic analysis, following Braun and Clarke (2006), a framework suitable to interpreting participants' narratives, analyzing and reporting patterns within the data. This method is appropriate for the study's aim as it facilitates and supports identifying meaningful patterns in how auditors describe their experiences of the interaction of skepticism and AI-generated outputs. As noted by Braun & Clarke (2006), thematic analysis provides the

opportunity to shift from description to interpretation, thereby revealing how meaning is formed throughout the dataset.

The analysis was based on the transcribed interview material. The analytical process adopted an iterative design, in which transcripts were examined repeatedly to gain familiarity with the material which is consistent with Braun & Clarke's (2006) logic. Multiple readings facilitated that later interpretations are anchored in a comprehensive understanding of the data. Subsequently, initial codes are created by marking segments that connect to the research focus such as how auditors articulate their evidence assessment, their comprehension of AI-generated outputs and how they perceive their skepticism exercise. These codes are reviewed and synthesized into broader themes that capture recurring patterns in which skepticism is exercised, or AI-generated outputs are relied on. The theme development process involves investigating relationships between various codes and whether they create a coherent and meaningful pattern. Themes are then assessed against the entire dataset to guarantee that they are distinct and empirically rooted in the participants' descriptions. This step is iterative, with themes being refined and adjusted as the study progresses, in line with Braun & Clarke's (2006) idea for theme refinement. The analysis involves both structuring the material and interpreting how participants make sense of AI-assisted audit settings. This enables the analysis to move beyond description towards themes that provide relevant analytical insights into how skepticism was expressed and at times, challenged when AI tools were involved.

Because thematic analysis involves interpretation, careful attention is given to conducting the process in a systematic and transparent manner. Coding is consistently applied across all the transcripts and interpretations are continuously checked against the empirical material. This helps to ensure methodological accuracy and transparency in how findings are reported throughout the analysis. An extract of the coding scheme is provided in Appendix 3 to demonstrate the development of codes and their connection to the themes.

## 3.5 Ethical considerations

Throughout the research process, ethical considerations are taken into account. Participation is entirely voluntary and all participants are provided with an information letter prior to the interview fulfilling the information requirement (David & Sutton, 2011). This letter specifies the study's aim, the nature of participation, the intended use of the collected material and the rights of the participants including the right to withdraw at any time without any explanation (see Appendix 2). Informed consent is acquired prior to each interview, in accordance with established ethical research principles (David & Sutton, 2011; Denscombe, 2017). Confidentiality is also maintained by avoiding the reporting of names or specific organizational affiliations. Details that could potentially expose a participant's identity are either removed or anonymized through the transcription and analysis phases. Upholding confidentiality is especially significant in qualitative research where participants share their experiences about their professional roles (Denscombe, 2017). All personal data, including interview recordings and transcripts are processed according to the General Data Protection Regulation guidelines and stored securely with limited access to the researchers and the purpose of the study (Regulation [EU] 2016/679 [General Data Protection

Regulation], 2016). After the completion of the study, the materials are managed with relevant data protection guidelines as highlighted in methodological literature (David & Sutton, 2011). Focus is also placed on how the questions are phrased, to prevent any pressure or discomfort. Creating an environment where respondents can freely express their point of view is crucial for ethical research (Denscombe, 2017). Thus, ethical considerations extend beyond formal procedures to guide the entire interactions during the data collection process.

## 3.6 Trustworthiness

Trustworthiness is a key consideration throughout the research process in order to ensure its quality. In qualitative research, trustworthiness is often framed in terms of credibility, dependability, and transparency rather than statistical validity (Denscombe 2017). Credibility focuses on whether the findings meaningfully represent the perspectives of the participants. In this study, credibility is strengthened through semi-structured interviews, which enable the participants to express their experience in their own language. The use of open-ended questions and follow-up probes facilitates more detailed responses, and the word-for-word transcription ensures that the analysis is rooted in the empirical data (Denscombe, 2017). Moreover, dependability is associated with the clarity and consistency of the research process (David & Sutton, 2011). To reinforce dependability, the study implements a systematic approach for data collection and analysis. The interview guide ensures that central topics are addressed constantly during the interviews, whereas the step-by-step process of thematic analysis offers a clear structure for interpreting the data which gives an understanding of how the findings were produced. Transparency is further promoted through detailed methodological choices and the rationale behind them. This included the participants' selection process, the data collection, the analysis process, as well as including empirical details and quotes from the participants. As Denscombe (2017) argues, making the research process explicit gives the reader the opportunity to assess the study's credibility on their own terms.

Furthermore, qualitative research involves interpretation as the researcher plays an active role in collecting and analyzing data. This introduces a degree of subjectivity which is acknowledged rather than eliminated. By maintaining a systematic analytical process and continuously grounding the interpretations in the data, the study seeks to minimize researcher influence. This involves re-evaluating coding choices, contrasting interpretations throughout the transcripts, and ensuring that the themes mirrored how the participants reasoned instead of the researchers' preconceptions. These steps help to ensure that the findings and conclusions remain tied and faithful to the participants' own accounts. Finally, this study does not aim for statistical generalization; instead the findings are intended to provide contextually grounded insights that may be analytically transferable to similar settings as Denscombe (2017) emphasizes that qualitative research prioritizes detailed and contextual understanding rather than broad generalization.

# 4 Findings

*This chapter presents the central patterns that emerged in the interviews and describes auditors' work with AI-based tools and how these impact their exercise of skepticism.*

## 4.1 Factors affecting the exercise of professional skepticism in the evaluation of AI-generated outputs

### *4.1.1 Professional experience and individual opinion*

One of the findings of the study is that professional experience constitutes a critical determinant in the exercise of professional skepticism in the evaluation of AI-generated outputs. The experience that the participants have is described as essential for assessing the plausibility of results, identifying anomalies, and determination of further investigation if required. During an interview, P4 said: *"It is that gut feeling I have...sometimes you can sense that something does not add up or in situations where you do not fully understand what is going on."* or P11 saying *"When you have worked with the same client for several years, you become very familiar with their operations. If something does not feel right, it is likely not right"*. Many participants further note that extensive experience develops an intuitive understanding of what is reasonable across varying audit contexts even when AI-generated outputs may appear accurate. Although the participants emphasize that AI can identify patterns and anomalies, in practice these results require human experience and a contextual understanding, which is difficult to replace or automate.

At the same time, there is a concern regarding less experienced auditors by the participants. It is argued by the respondents that junior auditors who often possess strong technical skills may find it more difficult to maintain a skeptical stance toward AI-generated outputs, particularly when structured. This was said by P10, *"But I believe we face a challenge when it comes to junior auditors, in that they tend to accept what AI says... sometimes without sufficient evaluation"*. The experience held by the auditors can act as a protective factor for professional skepticism, while a lack of experience may increase the risk of accepting AI outputs without sufficient critical assessment.

### *4.1.2 Audit methodology, standards and review procedures*

Alongside individual experience, audit methodology and established review routines are acknowledged as factors that affect the practice of professional skepticism. The participants indicate how methodology, standards, and formal requirements serve as frameworks that support and, in some cases, mandate a skeptical approach, even when utilizing AI-based tools. In particular the "four eyes" principle and the formal review processes are highlighted as essential mechanisms to ensure that AI-generated outputs are not accepted without critical evaluation as mentioned by P6: "*If an assistant has executed a task, it is essential for a manager to review it. This also applies when AI has performed a task; it must be reviewed by a human."* Or even as P1 exemplified it: *"In this context, the four-eyes principle is always in play, and if the initial two eyes belong to an AI while the final two are mine, then that is likely to be quite effective."*

The participants express that these routines contribute to the institutionalization of criticism and questioning within the audit process, rather than relying on the individual caution of auditors. Thus,

they describe the methodology as a safeguard against both habitual acceptance and unwarranted resistance to AI outputs. At the same time, a certain hesitancy emerges regarding standardized work programs and checklists. Some respondents view them as supportive of professional skepticism, while others argue that they may lead to a more mechanical work approach, where the focus shifts to fulfilling formal requirements rather than actively reflecting. As P4 articulates: *"Sometimes one also needs to lift one's gaze from merely following... checklist-based... to feeling I understand what I have done.*" Or as P5 illustrates: *"We may tick more items on various checklists but perhaps perform fewer actual actions... so one actually forgets what one is supposed to do."*. The participants show that methodology itself does not automatically guarantee professional skepticism, and the effectiveness of these structures instead seems to depend on how they are used and understood by the auditor.

### *4.1.3 The knowledge and understanding of AI systems*

Participants explain that the level of technical understanding of AI tools and in the assumptions embedded in their outputs shape both trust in and reliance of these tools. This further factor appears to increasingly influence the exercise of professional skepticism. A participant (P2) said: *"When it comes to how the result is generated, I want to understand 'how' ...that makes it easier for me to draw conclusions and decide what actions to take"*. Although this increases the level of skepticism among auditors, in parallel, uncertainty regarding the systems could limit the use of the systems even if they have the potential to contribute to the audit quality.

AI systems are also often perceived as complex, which increases the demands on auditors' ability to assess the reliability of the results. As P6 noted, *"The more you rely on the conclusion without fully understanding how it was reached, the more you need to question it"*. This finding indicates that a higher level of technical understanding facilitates a more balanced application of professional skepticism rather than uncritically accepting or dismissing it outright.

### *4.1.4 Organizational conditions: time pressure and workload*

The results reveal that organizational aspects including time constraints, efficiency expectations and workload have a substantial impact on the exercise of professional skepticism in AI-assisted audit settings. Participants describe how AI is typically introduced as a tool to enhance efficiency and reduce the time necessary for certain tasks in the audit process, as described by P6: *" There are such enormous time savings to be made"*.

Concurrently, efficiency objectives can in practice influence the extent of skepticism as explained by various participants. In contexts where workloads are considerable and the deadlines are tight, several participants describe that AI-generated outputs can easily facilitate advancing rapidly in their work, especially when no clear deviations are identified, as P5 illustrates: *"If you receive a report that is completely green, then it can be difficult, because where should I start digging... should I create problems that do not exist"*. P12 also states: *"I think stress is the major threat. If you are stressed... you become somewhat careless... and then you just move forward"*.

The participants highlight that this issue is not a matter of intentional neglect but rather a reflection of how work processes and incentives are designed. While AI has the capacity to allow more time

for analysis, some do experience that it tends to result in a quicker workflow rather than a more comprehensive assessment and review.

## 4.2 Professional skepticism as a continued human responsibility in AI-assisted auditing

### *4.2.1 Skepticism as a central part of the auditor's professional role and responsibility*

A clear and significant pattern in the data is that most respondents describe professional skepticism as a core aspect of the auditor's professional role. Skepticism is not mainly understood as a method or technical procedure, but rather as a foundational attitude that influences the entire audit engagement. Several participants express this in terms of responsibility, independence and professional identity. Participants highlight that the auditor's responsibility as the signing auditor remains unchanged, regardless of the advancement of the algorithmic tools employed. They believe that the ultimate responsibility for evaluations and conclusions can never be delegated to a system, for example as P2 said: *" I cannot protect myself by saying that AI said this or something like that, as I have 100% responsibility for everything I have signed.*" or as P11 indicates: *"In the end, it is a man or a woman who signs an audit report, certifying that everything appears correct and accurate. It is not an AI that does this.".* This viewpoint is prevalent and is often expressed clearly. Here, professional skepticism appears to be closely linked to the auditor's personal and legal duties, rather than the technical structure of the audit process.

Several participants articulate skepticism as the very essence of the auditing profession: *"You are an auditor and your role is to question and evaluate and so on, so I feel it is clearer than ever that this is what the job is"* (P7) or *"We are so ingrained in our profession to question everything and that includes AI"* (P3). In this light, skepticism emerges as a normative expectation and a professional ethical standard, rather than a task that can be reinforced or weakened by technology. Thereby, the utilization of AI does not transform the essence of being an auditor, but rather the context in which this responsibility must be carried out.

Simultaneously, it becomes clearer that AI tools in some situations increase awareness of this responsibility. As analytical tasks are automated and the growth of data sets, several respondents assert that human judgment becomes even more crucial. As the second participant (P2) points out: *"We will probably never get past the fact that we need a lot of people who think with their own heads and question things"*. Therefore, professional skepticism for the participants is characterized as a human counterpart to technical efficiency, serving as an essential check that guarantees auditing remains effective in fulfilling its purpose.

### *4.2.2 AI as a support but not a replacement for professional skepticism*

Across all interviews, AI is described as a support tool that enhances efficiency in parts of the audit process, especially in data collection, sampling and preliminary analysis. Simultaneously, the participants are clear that AI never gets used to independently making decisions or drawing conclusions. Numerous participants shared and used terms such as tools, input and assistance when

describing the function of AI in the audit process of their work, for example, P3 said: "*Ultimately it is always up to me to interpret the output and determine whether it is accurate. That is something the system will never be able to do without my involvement*" or as P1 mentioned: *"I see it 100% so far as a tool and as a support, and also that one can save time for more important matters."*. These outputs generated by AI are thus perceived as decision support that helps with the final audit opinion. AI can highlight patterns, anomalies or areas of risk, but these need to be situated in a context and assessed with the auditors' professional skepticism, according to the auditors.

Some of the participants emphasized that the approach does not change from how audit work performed by human colleagues is reviewed. For P12, "*You have to view these tools as an assistant, as if it were a human colleague, the same level of skepticism applies. You should not treat them as inherently better or worse*". This shows a pattern where AI in practice is treated the same as human resources while professional skepticism remains superior. At an operational level, AI seems useful, but participants underline that AI doesn't have the ability to account for contextual factors to the same extent as humans, such as organizational culture or subtle signals in client interactions. For example, P3 says "*We still need humans to make sense of it [AI output]. Okay, it has gotten this wrong for example. That part is where AI cannot assist us to the same extent*" or also P4 saying "*It provides the output, and it is up to you to analyze and understand them... you need to interpret the data and what it actually represents"*. Therefore, according to the respondents, the role of AI is limited to supporting the analytical phase rather than as a tool for independent decision-making.

### *4.2.3 Shifting the focus of skepticism: from the client's information to also the AI systems and their output*

Another subtheme highlights a distinct evolution in the focus of professional skepticism within an AI-assisted auditing framework. Traditionally, skepticism in auditing has been largely aimed at the information and explanations provided by the client. However, the data reveals that auditors are increasingly recognizing the importance of directing skepticism towards the AI tools employed in the audit and the results they yield. Several participants emphasize that AI outputs should not be accepted without critical examination and they must be assessed in terms of both content as P3 illustrated: *"It is so extremely important, after all, that as a user of AI you also check what output am I receiving",* and their origins as P6 pointed out *"If the result is as expected, but the path to it is wrong, then there is something that does not feel quite right"*. Therefore, skepticism now extends beyond questioning the accuracy of the client's information to also include the reasonableness of the AI system's assumptions, data sources and computational logic in the given context.

This transition is characterized by some participants as a new aspect of audit work, which participant 7 illustrates as follows:

> It becomes what I call our measures to verify the data that we call IPE in the audits, ..... we have six different risks that we usually go through and analyze, and it really doesn't differ that much. What differs with AI is probably that here we have an additional risk, meaning that we may add perhaps a seventh risk that the AI model could output something incorrect.

Thus, AI-generated outputs are considered as an additional layer of information that needs to be evaluated and questioned. The respondents argue that this raises the demands on the auditor's understanding of both the business and the technical tools.

Simultaneously, it becomes evident that this dual skepticism can be perceived as demanding. When AI tools are regarded as complex or opaque, the participants claim that the expectations on professional judgement increase rather than decrease: *"The more you rely on the conclusion, and you do not really know how the conclusion was reached, the more you have to ask questions"* as noted by P6. This evolution suggests that professional skepticism is not diminished in an AI-assisted audit context, but instead redefined and expanded. Skepticism is directed towards additional actors and sources of information, which strengthens the auditor's role as the ultimate guarantor of audit quality.

## 4.3 Challenges in maintaining professional skepticism in AI-assisted audit environments

### *4.3.1 Passive acceptance*

One of the prominent challenges identified in the material is the risk of passive acceptance of AI-generated outputs, especially when AI tools do not reveal any evident deviations. Participants describe how these systems are typically designed to flag risks or unusual patterns, which directs the auditor's attention towards what the system identifies as problematic. When no deviations are reported, the result often appears problem-free, which according to the respondents may reduce the incentive to conduct further examination as P5 notes: *"Nothing has been flagged, there are no deviations. Just sign it, .. But I think it's important that you always spend a certain amount of time per assignment and that you remain skeptical."* Or as P12 articulates: *"It can be easier to just accept the result. If you have more and more tools that just 'spit out' outputs, it can be easier to lose that [skepticism]."*

This situation is described as particularly problematic because the auditor's skepticism is potentially becoming reactive instead of proactive. Rather than autonomously questioning and assessing the results, the auditor becomes dependent on the system to identify the pertinent risks. Multiple respondents highlight that this issue is not only a matter of intentional blind faith in AI, but rather how the workflow transforms when AI is employed *"You get obviously help, and you might relax a little more"* describes P4. Several respondents also describe how AI output is often presented in a manner perceived as both logical and reliable, which may contribute to a reduced tendency to question the results. P6 states: *"One can be misled by the simplicity and neatness of the presentation, which may result in not completing the last step that should be taken, thus creating a trap that one might fall into"* supported by P10 that notes: *"It is often well-written, well-documented, says sensible things... then it really looks like one has thought things through"*.

The risk of passive acceptance, as expressed by the participants, is amplified by organizational factors such as time constraints and efficiency expectations which intensifies the tendency to blindly rely on the AI-generated outputs without further assessments.

### 4.3.2 Skepticism turning into limited use or resistance to AI

Alongside the risk for passive acceptance of AI-output shows an opposing dynamic in which professional skepticism manifests in the form of reluctance or resistance toward the use of AI-based analytical tools. Many respondents, especially more experienced auditors, describe skepticism as fundamental toward AI, which was rooted in uncertainty regarding algorithmic processes, the reliability of the outputs, and the assumptions they rely on. The auditors restrict their engagement with AI tools or assign less weight to its output, which evolves skepticism into a defensive approach: *"We are very... we rarely trust AI. If AI tells me something, it is usually not sufficient on its own ... I always have to go back to the source."* said P3. Certain participants explain that the resistance is the lack of interpretability of AI systems. When auditors are unable to understand how outputs are generated, they tend to adopt a default position of distrust. For example, P6 is one of the participants who said, *"You don't really know how the algorithm works, how it has generated or calculated its results... You simply have no insight into it. This is why (a) review (of the output) is so important"*

In these instances, skepticism appears less as an active evaluative process but rather as a general reluctance toward the technology. Several participants note that this may result in AI indications not being incorporated into the professional judgement, thereby limiting the depth of the audit. A citation by P6 during an interview:

> Let's say we miss something, and if we only used ChatGPT and did some kind of general scanning of the issue, we would have gotten the answer that we otherwise missed because we are only looking for papers, stamps and things of that nature.

This form of under-reliance is described as a challenge that may lead to inefficient use of AI tools rather than enhancing audit quality. According to the participants, this issue is associated with generational differences and varying levels of technological experience.

### 4.3.3 Tension between over- and under-reliance on AI

Overall, the findings show that the challenges perceived by auditors cannot be reduced to a simple question of trusting AI too much or too little. Instead of tension between two positions: a risk of passive acceptance of AI-generated output, and a risk that skepticism develops into resistance and under-reliance. This balancing issue is explicitly described by the participants and where professional skepticism requires continuous calibration of the use of AI. As said by P2: *"I think both involve significant risks...I believe the older auditors have a higher risk of not adopting it... whereas younger ones may not fully consider what they have actually input into the system"*. The participants emphasize that a balanced skeptical approach means that AI outputs should neither be accepted uncritically nor rejected in advance. Instead, the results should be used as a basis for further professional evaluation, where auditors actively weigh AI-generated indications against their own experience and contextual understanding. P7 took it to an extent and said, *"You maintain a reservation towards everything you see...you actually question things. That is what professional skepticism is."*

This calibration is recognized as a significant and demanding aspect of auditing within an AI-supported environment. Additionally, the participants mentioned that the difficulties stem from individuals experienced alongside organizational actors such as education and culture. Participant

6 said: *"Professional skepticism is a matter of culture that should be embedded in all parts of the organization...it is about attitudes and how carefully you perform tasks"*. An audit environment where critical questioning is encouraged and where the application of AI is followed by reflection and discussion is highlighted as important. Further, P10 showed the importance of professional skepticism when said: *"The whole profession is built on it....we are trained in professional skepticism because it is a fundamental principle of the profession."*. Therefore, these findings indicate that professional skepticism remains a core principle, but that auditors face increasing demands in balancing the reliance on and distance from AI-generated outputs.

# 5 Analysis

*The chapter analyzes and discusses the results by integrating them with prior research on AI, skepticism and cognitive mechanisms to explain how and why skepticism is reshaped in AI-driven audit settings.*

## 5.1 Factors affecting the exercise of professional skepticism in the evaluation of AI-generated outputs

The study's findings demonstrate that auditors' application of professional skepticism in AI-assisted auditing is influenced by a dynamic interaction of experiential, methodological, technical and organizational factors. These factors do not function in isolation; instead, they collaborate to affect how auditors interpret AI-generated outputs and assess whether additional audit procedures are necessary. This reflects Nelson's framework (2009), which suggests that skeptical judgment and action are influenced by contextual cues, evidential signals, and auditors' cognitive assessments. Consistent with the theoretical synthesis, the findings show that AI reshapes both the evidential environment and the behavioral conditions under which skepticism is enacted.

Professional experience plays a crucial role as an interpretive lens in how auditors exercise skepticism when evaluating AI-generated outputs. Auditors that are experienced can draw on industry knowledge and familiarity, derived from prior engagements with the clients' information patterns. By that means, experience enables the auditors to evaluate whether the AI outputs are helpful or relevant in the audit context. This supports Nelson's (2009) argument that auditors' knowledge and experience shape their ability to construe evidential cues and decide if further assessment is needful. In alignment with ISA 315, where central to risk assessment is the understanding of entity and the environment it is in. The AI-generated outputs only possess audit meaning with contextual interpretation in relation to client context, audit risk and materiality. Brown-Liburd et al. (2015) additionally demonstrate that AI-based analytical tools mold how auditors evaluate information. The findings reveal a contrast between skeptical action and skeptical mindset. Although AI may prompt more questioning, especially among junior auditors, it does not necessarily translate into deeper investigative procedures showing the experience's dual role. However, a novel insight emerging from the findings is that the automation of routine tasks through AI; may constrain junior auditors' opportunities to gain experiential basis needed for future skeptical evaluations. This contributes the existing skepticism literature by indicating that AI not only influences skepticism within individual audit engagements but reshapes how the skeptical competence develops over time within the profession.

Methodological frameworks, including review procedures and the four-eyes principle, act as institutionalized mechanisms designed to enhance skeptical actions. Consistent with ISA 200 and ISA 315, these structures are established to guarantee that AI-generated outputs are not accepted without critical evaluation and that contradictory evidence is actively considered. Yet, findings also reveal a well-documented risk in professional skepticism literature: structured audit tools may facilitate procedural compliance without necessarily strengthening the auditor's fundamental judgment. Nelson's (2009) differentiation between a skeptical judgment and skeptical action is crucial in this context. While checklists and standardized review procedures may encourage auditors to adopt a skeptical mindset, they do not ensure the active evaluative efforts needed to transform

that judgment into substantial skeptical action. This aligns with Nolder and Kadous' (2018) argument that skepticism requires active interaction and engagement with evidence rather than reliance on predefined steps or check the box approaches. When structure becomes the main focus, auditors risk conducting the audit "correctly" following the procedures, without critically examining the evidential foundation of their conclusions.

This tension is amplified in AI contexts. The structured and nature of AI-generated outputs can foster a misleading sense of confidence, leading auditors to feel that the analytical work has already been performed. This may result in a situation where a skeptical mindset is present, but the skeptical action is weakened; a discrepancy that still not adequately addressed in traditional frameworks. Thereby, methodology and audit practices can simultaneously support and unintentionally compromise skeptical engagement depending on whether auditors apply it as cognitive support or a procedural shortcut.

The role of technical understanding further shapes the ability of auditors to critically evaluate algorithmic outputs. The findings show that auditors need a practical comprehension of how the inputs and model assumptions influence the outputs they receive which does not require the auditors to become technical experts, but to have a sufficient AI knowledge and understanding. This is in line with prior research suggesting that transparency and perceived control over analytical tools influence auditors' trust behaviors (Commerford et al., 2022; Li et al., 2026). Without this understanding, auditors risk misinterpreting outputs by either overestimating or undervaluing them, extending the literature by showing that technical understanding affects also how auditors direct their skepticism. Thus, AI literacy becomes an integral component of skeptical judgement extending the traditional definitions of professional competence. This is consistent with the theoretical synthesis which argues that skepticism in AI-assisted auditing involves assessing not only the evidential output but also the mechanism through which it is generated. A further contribution of the findings is that AI literacy appears to function as a mechanism for rigorously evaluating AI-generated outputs and a moderating factor between simply accepting them or rejecting them; suggesting that technical understanding is not merely supportive but foundational for calibrated skepticism.

Organizational conditions, particularly time pressure and workload, emerge as important factors that shapes the exercise of professional skepticism in AI-assisted environment. While AI is often presented as a tool that frees time to allocate more time to analytical evaluation, the findings instead indicate a paradoxical effect. Even though it leads to faster workflows, that may encourage more passive acceptance of AI generated outputs instead of deeper critical assessment before acceptance. This dynamic supports research on cognitive miserliness and increased reliance where individuals under time pressure rely on cognitive shortcuts such as automated recommendations rather than independent deep evaluations (Skitka et al., 1999). A notable finding is that organizational incentives emphasizing efficiency and productivity may unintentionally reinforce auditors' trust in AI-generated outputs despite maintaining a skeptical attitude in principle. This resonates with the theoretical synthesis that skepticism is also shaped by organizational pressures, algorithmic signals and behavioral tendencies. Consequently, AI introduces a structural tension whereby it has the potential to enhance audit quality by assisting auditors in managing large volumes of data and

identifying potential high-risk areas, which corresponds with AI research (Appelbaum et al., 2017; Barr-Pulliam et al., 2024; Kokina & Davenport, 2017); yet it simultaneously reduces the level of critical scrutiny applied in practice. This indicates that the impact of AI on professional skepticism relies on the AI's integration within the audit work environment rather than solely the system's technical capabilities.

Overall, the findings suggest the enactment of professional skepticism in AI-assisted auditing is the result of a relationship between auditor experience, institutionalized audit procedures, technical understanding and the organizational pressures. Experience gives context, methodology adds safeguards, technical understanding enables outputs evaluation and organizational conditions impact if real evaluation is possible. These factors may reinforce or conflict each other, for example an auditor may have strong experience but low AI knowledge, or a firm may have formal procedures while time pressure undermines them. The analysis is consistent with the theoretical synthesis, demonstrating that AI changes both the character of evidential cues and the behavioral conditions under which auditors interpret them. Furthermore, the findings extend existing literature by indicating that AI may shape the evolution of skeptical competence over time, thereby introducing an emergent perspective to the existing in audit digital context. This suggests that skepticism in AI-assisted auditing is optimally understood as a dual process: interpreting algorithmic signals and managing the behavioral and structural tendencies that shape auditors' responses to those signals.

## 5.2 Implications of AI for auditors' skeptical responsibilities

The findings show that auditors perceive professional skepticism as a continually central and non–delegable part of the auditor's professional role. This can be understood in relation to ISA 200, where professional skepticism is defined as a questioning mindset and critical assessment of whether audit evidence is sufficient and appropriate (IAASB, 2009). While traditional professional skepticism literature assigns the auditor the responsibility of critically assessing audit evidence rather than just receiving or processing information (Nelson, 2009; Nolder & Kadous, 2018); the findings indicate that AI alters the type of evidence to which this responsibility pertains. However, the findings extend this literature by showing that, in AI-assisted auditing, the scope of responsibility expands beyond client information and traditional audit evidence to include these AI generated outputs and the processes that produce them.

Furthermore, these findings are consistent with prior research on the transformation of the audit evidence environment. Appelbaum et al. (2017), Barr-Pulliam et al. (2024), and Kokina and Davenport (2017) emphasize that AI alter both the production and evaluation of audit evidence. The present findings corroborate this view, as auditors conceptualized AI-generated outputs such as anomaly flags, summaries or risk indicators as integral components of the evidential base. Simultaneously, the findings contrast and complicate this literature by challenging a purely efficiency-driven understanding of AI in auditing. While AI improves analytical capacity and pattern detection, it reconfigures the exercise of professional skepticism by making it more indirect and layered evaluative process.

The findings further align with Gu et al.'s (2024) notion of co-piloted auditing, where AI supports auditors' decision-making instead of replacing the professional skepticism. Likewise, Lehner et al. (2022), Munoko et al. (2020) and Murikah et al. (2024) argue that AI can enhance audit work, yet it also raises concerns about the necessity of human oversight, accountability and ethical responsibility. The findings support this viewpoint as participants described AI as useful for data processing, summarization, anomaly identification and analytical review while simultaneously emphasizing that final evaluative responsibility remains with the auditor. Nevertheless, the findings introduced an important nuance by showing that AI does not operate as a neutral support tool, but it actively structures the informational environment by influencing which anomalies, patterns and risks are made salient to the auditors. This resonated with Brown-Liburd et al. (2015), who argue that AI-based analytical tools influence how auditors interpret and prioritize information. As a result, despite the fact that auditors perceive AI as a form of support, it shapes the circumstances in which professional skepticism is exercised, as the evidential environment has already been partially filtered and framed through algorithmic processing.

Furthermore, the results are consistent with Nolder and Kadous' (2018) framework, as auditors demonstrate a questioning stance toward AI-generated outputs while reinforcing that professional accountability cannot be transferred to technological systems. However, the findings also demonstrate that active engagement becomes more demanding and complex when audit evidence is produced through algorithmic systems. This is because auditors must not only interact with the visible content of the output, but also assess the validity of inputs, underlying assumptions and limitations. This underscores an important conceptual gap between skepticism as attitude and as evaluative practice; because AI-assisted auditing requires auditors not only to be cautious toward AI but also to interrogate how technological processing has shaped the evidence.

Li et al. (2026) argue that transparency and documentation influence auditors' skeptical evaluation of analytical tools. The findings correspond with this perspective, as auditors' ability to exercise professional skepticism and assume responsibility for audit conclusions is closely tied to their understanding of how AI-generated outputs are produced. This also aligns with Bertomeu et al. (2021), Parker et al. (2025), and Rahman et al. (2024), who emphasize that algorithmic systems can create challenges related to explainability, accountability and trust in professional decision-making. The findings resonate with this research stream by demonstrating that auditors' ability to uphold accountability relies on their understanding of the generation process of the outputs. However, the findings also advance existing literature by indicating that transparency functions not merely as technical determinant of reliance, but as a prerequisite for meaningful accountability in AI-assisted auditing. In situations where auditors remain formally accountable for audit conclusions, limited insight into the underlying logic, assumptions and processing mechanism of the AI systems constrains their capacity to critically evaluate the reliability and evidential significance of system-generated outputs.

It emerges in this study that AI does not undermine auditors' skeptical responsibilities; instead, it broadens their scope from direct evidence evaluation to assessing system-mediated evidence. This is where the theoretical synthesis becomes particularly evident as AI-generated outputs function as algorithmic signals within the audit evidence environment, which need to be verified, interpreted

and contextualized to be audit-relevant. In consistency with the synthesis, professional skepticism emerges as the balancing mechanism through which auditors assess and respond to these signals. However, the data deepen the literature by demonstrating that this balancing function depends on auditors' ability to assess critically both the evidential content of the output but also toward the logic and technological process through which such outputs are constructed.

## 5.3 Challenges in maintaining professional skepticism in AI-assisted audit environments

A central insight from the findings is that the exercise of professional skepticism in AI-assisted auditing is less about trust in technology and more about the calibration of reliance on algorithmic outputs. This aligns with behavioral research showing a systematic tension between over-reliance on automated systems and algorithm aversion (Dietvorst et al. 2015; Logg et al., 2019; Parasuraman & Riley, 1997; Skitka et al., 1999). Within auditing, this duality tension becomes significant because professional skepticism requires auditors to critically evaluate AI-generated outputs without accepting them passively or dismissing them prematurely. The findings therefore suggest that the core challenge lies in operationalizing skepticism as a calibrated evaluative process.

One challenge identified is the passive acceptance of AI-generated outputs. The participants described a tendency that appears in situations where outputs get interpreted as sufficient when they appear structured, complete, logical, or when no anomalies get flagged by the system. This observation is aligned with research on automation bias, which demonstrates that users tend to reduce independent evaluation when automated systems appear reliable (Parasuraman & Riley, 1997; Skitka et al., 1999). However, the findings identifies an audit relevant mechanism whereby the risk is not only that the auditors follow and show compliance to incorrect AI recommendations, but that they fail to investigate areas that AI has not highlighted, aligning with Skitka et al.'s (1999) and Romeo and Conti's (2026) identification of omission errors as primary type of automation bias. In this sense, automation bias emerges as an omission risk, whereby the absence of algorithmic shapes the boundaries of what auditors perceive as requiring further examination.

This issue is crucial in auditing as the absence of AI-generated warnings does not automatically indicate that audit risk is absent. According to ISA 200, auditors must rigorously assess whether the evidence is both sufficient and appropriate, while ISA 240 underscores the necessity for maintaining professional skepticism as fraudulent misstatements may be deliberately hidden (IAASB, 2009). The findings reinforce this concern, as auditors may interpret absence of anomaly detection as an implicit confirmation of low-risk area, causing their skepticism shift from active questioning to passive acceptance. This extends the literature by demonstrating how omission risk can become relevant to audits when system silence is perceived as evidential reassurance.

Algorithm appreciation is useful here as well as a supportive mechanism to the over-reliance issue. Research by Logg et al. (2019) and Parasuraman and Riley (1997) demonstrate that algorithmic recommendations may be trusted more because they seem objective, analytical and sophisticated. This is supported by the findings as auditors find that well-presented and structured AI-outputs to be reassuring. However, the audit-specific contrast reveals that a credible presentation does not

necessarily guarantee the output's relevance or sufficiency as its value is still dependent on auditors' interpretation, the quality of data, underlying assumptions and consistency with other evidence.

A contrasting challenge identified is the resistance towards AI-generated outputs and the conditions under which auditors become reluctant to incorporate such outputs into evidential assessments. For example, the findings show that when AI output is difficult to verify or understand within the audit context, the auditors may become cautious to it. This aligns with the literature on algorithm aversion, which argues that individuals may discount the algorithmic recommendations in situations when systems appear inflexible or insufficiently transparent (Dietvorst et al., 2015). Furthermore, alignment with research showing that reliance on analytical tools is structurally shaped by the task complexity, perceived control, system flexibility and auditor involvement (Commerford et al., 2022; Ulla & Commerford, 2025). However, the findings also nuance this literature by showing that resistance or reluctance toward AI-generated outputs in auditing is not necessarily indicative of irrational distrust. Within auditing, such caution towards AI may reflect legitime professional skepticism when outputs are opaque or difficult to reconcile with other audit evidence.

The indications in findings illustrated that difficulties arise when evaluative caution is transformed into premature dismissal of AI-generated outputs. Nolder and Kadous' (2018) distinction between questioning mindset and active engagement clarifies this dynamic, as skepticism requires not only critical thinking but also its translation into investigative action. Meanwhile, AI may trigger a questioning mindset through e.g. anomaly flags or risk indicators, this does not necessarily result in active engagement in the form of further audit procedures. Thus, under-reliance becomes problematic when auditors reject these signals generated by AI without any consideration on whether they will provide any relevant audit insight. This finding contributes to the professional skepticism literature by revealing that skepticism in AI-assisted auditing can be weakened not only by over-trust/reliance in automation, but also by failure to convert the critical awareness into active engagement and investigative action.

Calibrated trust is suggested by the findings as the main challenge. Auditors did not explain AI as either trustworthy or unreliable, instead they emphasized situating algorithmic outputs within broader evaluative criteria such as audit risk, materiality, task complexity and data quality. This reflects ISA 200's (IAASB, 2009a) requirement to evaluate the sufficiency and appropriateness of audit evidence, and Nelson's (2009) skeptical action, where auditors' evaluative assessment precedes the decision on whether the output should be verified, challenged or supplemented with further examination through audit procedures. Consequently, calibrated trust reconceptualizes skepticism as a procedural mechanism for regulating reliance on AI-generated outputs rather than a general attitude.

Similarly, Li et al. (2026) support this interpretation and argues that transparency and documentation influence auditors' skeptical evaluation of AI tools. The findings align with this literature, yet they highlight that calibrated trust cannot be solely derived from transparency. The findings extended the prior literature by illustrating that transparency is not only associated with reliance decisions, but the auditors need to relate the AI-output to audit risk, materiality and other

evidence. This is essential as an AI output may be clear yet insufficient and advanced yet irrelevant. Calibrated trust is therefore necessitating that auditors assess both the technical validity and the audit significance of the outputs

Altogether, the findings demonstrate the main challenge in maintaining professional skepticism in AI-assisted auditing is navigating between automation bias and algorithmic aversion. This operationalizes the theoretical synthesis by demonstrating how auditors' behavioral responses to algorithmic signals shape whether skepticism functions effectively as a balancing mechanism. Automation bias may reduce the skeptical evaluation by encouraging unreflective acceptance of system-generated outputs, whereas algorithm aversion may result in the premature rejection of potentially valuable insights. The findings further contribute to the literature by suggesting that professional skepticism increasingly functions as form of calibrated trust, where auditors continuously assess the reliability, limitations and contextual relevance of AI-generated information. Furthermore, skepticism becomes less about generalized distrust and more about the auditor's capacity to critically evaluate when algorithmic outputs are reliable or require further investigative procedures.

# 6 Conclusion

*This concluding chapter summarizes the study's central insights and discusses its implications, identifies imitations and suggests directions for future research.*

The purpose of this study is to examine auditors' perceptions of exercising professional skepticism when evaluating AI-generated outputs as part of audit evidence. To achieve this purpose, the study is guided by three research questions. These address auditors' perceptions of the factors influencing their exercise of skepticism in AI-assisted audit settings, the implications of AI for their skeptical responsibilities and the challenges of maintaining professional skepticism when evaluating AI-generated outputs.

The study's key concludes that the practice of professional skepticism is affected by multiple factors. Professional experience plays a crucial role as it enables auditors to make reasonableness assessments and determine when AI-generated outputs require additional review. While audit methodologies, review processes and the "four-eyes" principle support critical analysis, they do not guarantee skepticism is exercised mechanically. Additionally, a solid technical understanding is identified as an important prerequisite, as auditors need to comprehend the generation of AI outputs to evaluate their reliability. The study indicates that organizational factors, such as time pressure and efficiency requirements, impact the auditor's ability to actually apply a skeptical approach in audit procedures.

The study furthermore concludes that professional skepticism is still perceived as a human and non-delegable responsibility in AI-assisted auditing. Auditors describe AI as a supportive tool that can facilitate and optimize various phases of the audit process, for instance by organizing and reviewing data, identifying anomalies or summarizing information. Nevertheless, AI is not considered capable of replacing the auditor's professional skepticism and overall evaluative judgment. The auditor remains accountable for interpreting, verifying and assessing AI-generated outcomes in relation to other audit evidence. The study also shows that AI is altering the evidential context within auditing. Auditors are now confronted with not only the original client data but also with system-generated representations of that data. This necessitates that professional skepticism be applied not solely to the client's information but also to the outputs, data sources, and underlying assumptions of AI tools. As a result the realm of skepticism expands as AI becomes incorporated into the auditing process.

A significant conclusion is that auditors need to find a balance between over-reliance and under-reliance on AI. The risk of excessive trust emerges particularly when AI-generated outputs seem logical, structured or when no anomalies or deviations are flagged. This can result in a more passive review process. On the other hand, there is an opposing challenge where skepticism may turn into resistance or limited use of AI, especially when tools are perceived as obscure or difficult to comprehend. Consequently, the study indicates that professional skepticism in AI-assisted auditing should be a calibration practice as auditors should neither accept AI-generated outputs uncritically nor dismiss them outright, but evaluate them appropriately in relation to risk, context and other audit evidence.

## 6.1 The study's contributions and implications

The study contributes by deepening the understanding of auditors' perceptions of professional skepticism in AI-assisted auditing settings. AI not only affects efficiency and analytical performance but also alters the nature of what auditors need to be skeptical about. An important contribution is how the objects of skepticism expand from the information by the client to include AI-generated outputs and the systems behind them. The study also contributes by linking professional skepticism with automation bias and algorithm aversion, highlighting how excessive trust or distrust may impair auditors' ability to appropriately evaluate audit evidence.

Building on these insights, the study also offers practical implications for audit practice. A need emerges for audit firms to develop clear guidelines for how AI-generated outputs should be used, verified and documented. AI output is not and should not be treated as final audit evidence but rather as analytical input that still requires professional evaluation. Therefore, training in AI should focus on how auditors critically examine the AI systems' data quality and their output. Additionally, the study indicates that review processes should not only cover conclusions but also how they are produced. To ensure that time saved through AI is redirected from efficiency-driven use toward deeper reflection and analysis.

## 6.2 Limitations and future research

As with any qualitative research, certain limitations should be acknowledged. The study is based on a limited number of interviews with authorized auditors and therefore cannot be statistically generalized to the auditing profession as a whole. Even though the data have nuanced and rich insights into how auditors perceive professional skepticism in AI-assisted auditing environments. Furthermore, the study relied on auditors' own perceptions that were reported rather than on observation of actual audit practice. There may thus be a gap between how auditors describe their skeptical approach and how they act in real-life situations. Additionally, even though the participants were authorized auditors from Big 4 firms with different lengths of experience within the auditing industry but also varying levels of experience with AI tools. This is a limitation since certain participants described AI as integrated part of the daily work while others described a more limited use. This could create a variation in the data that could be analytically valuable but also means that the results reflect different levels of maturity in AI. Hence, it can be difficult to draw any clear conclusions about AI's impact on professional skepticism in practice.

However, these limitations unfold possibilities for further research. The studies in the future could use observational design or experiments to examine how auditors assess AI-generated outputs in practical situations. Also comparing senior auditors with junior auditors, in order to analyze whether and how automation bias and algorithm aversion differ across experience levels. Further studies might also examine the effects on long-term development of professional skepticism, as well as how auditors' training and skill development need to evolve as AI becomes more prevalent in audit practices. Moreover, research could compare various types AI tools, to assess whether the challenges of maintaining professional skepticism vary depending on the complexity, transparency and purpose of the system in question.

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# Appendix 1- Interview guide

Introductory questions

1.Can you briefly describe your role and experience in auditing?
   - How long have you worked as an auditor?
   - How long have you been a certified (authorized) auditor?
   - What types of audit engagements do you primarily work with?

Use of AI

2. To what extent are results from AI-based analytical systems used in the audit work where you work?
   - In which parts of the audit process are they used?
   - Can you provide examples of the types of analyses or signals the systems generate?
3. How do you perceive the role of such system-generated indicators in audit work?
   - Are they primarily used to support planning, risk assessment, or evidence evaluation?
   - How do you perceive that they impact your work in practice?

Professional skepticism

4. How would you describe what professional skepticism means in your day-to-day work?
   - How does it manifest in practice?
   - In which situations is it particularly important?
5. How do you view your role in maintaining a critical and questioning approach when evaluating audit evidence?
   - What might lead you to question something?
   - Are there specific practices or routines that assist you in this?

Evaluation of system-generated outputs/analyses

6. When outputs or analyses from such systems are used as part of the audit evidence, how do you approach interpreting and evaluating them?
   - What determines whether you consider such output reliable or relevant?
   - How do you relate these outputs to other audit evidence?
7. How do you perceive such AI-generated outputs in relation to traditional audit evidence?
   - Do you see them as equivalent or different?
   - Are there differences in how you interpret or assess them?
8. How do you work to critically review and verify AI-generated analyses before they are used as a basis in the audit?
   - Do the results often need to be verified with other sources?
   - Are there situations where you choose to examine the results further?

Balancing between AI output and own analysis

9. How do you perceive the balance between analyses from AI tools and your own analysis of audit evidence?
   - Are there situations where the system's analysis carries more or less weight?
   - What affects how much importance you place on the system's results?
10. Can you describe a situation where a system-generated analysis/result influenced how you evaluated audit evidence?
   - How did you handle that situation?
   - Did it affect how you evaluated the audit evidence/How did it affect your subsequent analysis?

Challenges and opportunities

11. Do you experience situations where there is a risk of relying too much or too little on AI-generated results?
   - Can you give examples of such situations?
   - What influences how much you choose to rely on the results (e.g., your experience with the system, the type of task, or how the system operates)?

12. How do you work to ensure that you do not rely too much or too little on such systems?
13. How do you perceive challenges in maintaining professional skepticism when AI tools are used in audit work?
    - Can such systems influence how evidence is interpreted?
    - Can you provide examples of situations where the system's analyses have affected how you interpreted or assessed audit evidence?
    - Are there risks associated with relying on the system's output, and if so, what are they?
14. Are there factors that help you maintain a critical and independent approach when using AI tools? (e.g., experience, methodology, and discussions within the audit team)

    Closing questions
15. How do you perceive your responsibility when system-generated results are used as part of audit evidence?
16. Has your approach to working with skepticism changed as such analyses are used more frequently?
17. How do you think the use of such analytical systems may affect audit work in the future?
    - Do you think it could influence how professional skepticism is exercised?
    - How should the audit profession handle such changes?
18. Is there anything in your work with AI-generated output that you believe is particularly important to understand regarding how auditors deal with it?

# Appendix 2- Information Letter

**AI Assisted Auditing and Professional Skepticism: Auditors' Perceptions of AI-Generated Audit Evidence**

Thank you for agreeing to participate in our study. We are currently in our final semester of the Business and Economics program at Karlstad University and are writing our Degree Project. Below, you will find more information about the study and what participation entails.

The purpose of our thesis is to examine how auditors perceive and exercise professional skepticism in audit environments where AI-generated outputs are used as part of the audit evidence. The focus is on how auditors view the evaluation of such outputs during audit engagements. We are therefore seeking auditors to interview on this topic. The interview will be conducted either in person or remotely via Zoom or a similar platform and it is expected to take approximately 20–40 minutes. The interview will be recorded and later transcribed into written form.

Your participation in the study is voluntary, and you may withdraw at any time. If you choose not to participate or decide to discontinue your participation, you do not need to provide a reason. If you wish to withdraw, please contact the researchers responsible for the study (see below). The study will collect and store information about you. Your responses and results will be handled in a way that prevents unauthorized access. The data will be retained until the thesis is completed and will then be destroyed. Your name or any other directly identifying information will not appear in the results. The legal basis for processing your personal data is informed consent. You may withdraw your consent at any time without providing a reason.

Karlstad University is the data controller for the processing of your personal data. The data may also be processed by data processors, such as cloud service providers. Under the EU General Data Protection Regulation (GDPR), you have the right to access the information about you that is processed in the study, and to have any inaccuracies corrected. You may also request that your data be deleted or that the processing of your personal data be restricted. If you wish to access your data, please contact the researchers responsible for the study (see below). If you are dissatisfied with how your personal data is handled, you have the right to lodge a complaint with the Swedish Authority for Privacy Protection (Integritetsskyddsmyndigheten), https://www.imy.se , which is the supervisory authority. Contact details for the Data Protection Officer at Karlstad University are: email dpo@kau.se , telephone (switchboard) +46 (0)54 700 10 00. For more information on how Karlstad University processes personal data, see https://www.kau.se/gdpr.

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## Appendix 3- Extract from the coding scheme

| Theme | Category/subtheme | Codes | Data example |
|---|---|---|---|
| Skepticism: Continuous human responsibility | Skepticism as a professional role/responsibility | Responsibility, independence, signing, professional role, question, evaluate, think | P2: I cannot protect myself by saying that AI said this or something like that, as I have 100% responsibility for everything I have signed.<br><br>P1: It always needs to be a person behind it who takes responsibility for things.<br><br>P2: It's like there's always a signing auditor who is responsible for everything.<br><br>P3: we are so ingrained in our profession to question everything and that includes AI.<br><br>P5: you still have a responsibility to find things that we are supposed to discover.<br><br>P7: your role is to question and evaluate and so on, so I feel it is clearer than ever that this is what the job is.<br><br>P7: I am still responsible for it regardless; it changes nothing. It is ultimately my responsibility to analyze and document.<br><br>P12: as long as the human signs the audit report, it is the human who is responsible. |
| | AI as aid not replacement | tools, aid, assistant, support | P1: I see it 100% so far as an aid and as a support.<br><br>P1: I see it as a great aid to be able to do some of the work before you yourself jump in and review or analyze or draw conclusions.<br><br>P9: So far it has been an aid. |